

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	30 September 2026
Team ID	SWTID-2026-1429
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Model Selection & Architecture	USN-1	As a developer, I can research and compare Gemini Flash, Gemini Pro and Stable Diffusion so that the best models are selected for outline, story and image generation.	5	High	Sanjaykavin S, Mohamed Bowmin R
Sprint-1		USN-2	As a developer, I can define the architecture (Frontend, FastAPI Backend, AI Integration) so that the data flow of the application is clear.	3	High	Sanjaykavin S, Mohamed Bowmin R, Sakthivel S, Soniya N
Sprint-1		USN-3	As a developer, I can set up the virtual environment, install dependencies and create the project structure so that the FastAPI server runs locally.	2	High	Sakthivel S
Sprint-1	Core Functionality Development	USN-4	As a user, I can enter a story prompt and receive a structured 5-panel comic outline generated by Gemini Flash (generate_outline()).	5	High	Sanjaykavin S
Sprint-1		USN-5	As a user, I can get detailed narration and character dialogue for every panel generated by Gemini Pro (generate_story()).	5	High	Mohamed Bowmin R

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-2		USN-6	As a user, I can get a comic-style illustration for each panel generated by Stable Diffusion (generate_image()).	5	High	Soniya N
Sprint-2		USN-7	As a user, I can see each panel's image matched with its story text through the layout builder (build_comic_layout()).	3	High	Sakthivel S
Sprint-2		USN-8	As a user, I can export my comic to a multi-page PDF using FPDF (save_pdf()) so that I can save, print or share it.	5	High	Sanjaykavin S
Sprint-2	Backend & Routing	USN-9	As a developer, I can implement the FastAPI application and routing configuration so that user input is processed by the backend.	4	High	Mohamed Bowmin R
Sprint-2		USN-10	As a user, I can submit the form or a JSON API request in routes.py and receive a generated comic.	3	High	Soniya N
Sprint-3		USN-11	As a user, I can use the image-testing and export-confirmation routes so that generation and export are verified.	3	Medium	Sakthivel S
Sprint-3	Frontend Development	USN-12	As a user, I can enter story prompt, character name, setting, tone and art style in a clean input form (index.html).	5	High	Sanjaykavin S
Sprint-3		USN-13	As a user, I can preview the generated comic panel by panel on the web page (comic_preview.html).	5	High	Soniya N
Sprint-3		USN-14	As a user, I am redirected to a success page after downloading my comic (export_success.html).	3	Medium	Mohamed Bowmin R
Sprint-3		USN-15	As a user, I get a responsive, well-styled interface built with HTML, CSS and dynamic Jinja2 templates.	4	Medium	Sakthivel S
Sprint-4	Deployment & Testing	USN-16	As a developer, I can prepare the application for local deployment (requirements file, API key configuration, folders for panels and exports).	5	High	Sanjaykavin S
Sprint-4		USN-17	As a user, I can change tone (e.g., funny) and art style (e.g., comic book) and regenerate the whole comic to match my vision.	5	High	Mohamed Bowmin R

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-4		USN-18	As a tester, I can run end-to-end tests and verify local deployment at http://127.0.0.1:8000 including PDF export.	5	High	Soniya N, Sakthivel S
Sprint-4		USN-19	As a team, we fix bugs and finalise project documentation for submission.	5	Medium	Sanjaykavin S, Mohamed Bowmin R, Sakthivel S, Soniya N

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	05 Oct 2026	10 Oct 2026		
Sprint-2	20	6 Days	12 Oct 2026	17 Oct 2026		
Sprint-3	20	6 Days	19 Oct 2026	24 Oct 2026		
Sprint-4	20	6 Days	26 Oct 2026	31 Oct 2026		

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\textit{sprint duration}}{\textit{velocity}} = \frac{20}{10} = 2$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>
<https://www.atlassian.com/agile/tutorials/burndown-charts>

Reference:

<https://www.atlassian.com/agile/project-management>
<https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software>
<https://www.atlassian.com/agile/tutorials/epics>
<https://www.atlassian.com/agile/tutorials/sprints>
<https://www.atlassian.com/agile/project-management/estimation>
<https://www.atlassian.com/agile/tutorials/burndown-charts>